



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	RAZIA BANU
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Roll Number	2520030534
Branch	CSE
Course Outcome	CO3

Case Study Topic: Airline Connectivity Analysis using DSU Rollback (Set 9 Q3)

Work Done Summary:

In this case study, a **Disjoint Set Union (DSU) with Rollback** data structure was implemented to analyze connectivity within an airline transportation network. Airports were modeled as vertices in a graph, while flight routes between airports were represented as edges connecting those vertices. The DSU structure was used to efficiently determine whether two airports belonged to the same connected component, enabling quick connectivity checks across the network. To optimize performance, the **Union-by-Rank** technique was applied during merge operations, ensuring that the resulting trees remained balanced and reducing the overall complexity of union and find operations.

A rollback mechanism was incorporated to support dynamic changes in the network. This feature allowed the system to restore previous graph states after processing individual queries, making it possible to evaluate the impact of adding or removing flight routes without permanently altering the network structure. Connectivity results were analyzed and compared before and after route modifications to observe how network accessibility changed over time. The implementation demonstrated that DSU with Rollback is an efficient and practical solution for handling **dynamic connectivity problems**, where connections frequently change and historical states need to be revisited. Such an approach is highly applicable in airline management systems, transportation networks, communication systems, and other real-world scenarios that require fast and reliable connectivity analysis under changing conditions.

Implementation Details:

1. Implemented parent and rank arrays.
2. Applied union-by-rank.
3. Stored rollback history.
4. Restored graph state after rollback.
5. Compared with BFS/DFS approach.

Result: Screenshots / Output

```
PS D:\tmp> cd "d:\Coding\Projects\Case Studies" ; javac DSURollbackConnectivity.java ; java DSURollbackConnectivity
AIRLINE CONNECTIVITY ANALYSIS

Creating Airline Routes...

Connectivity Check
Connected (0,3): true

Performing Rollback...

Connected (0,3): false

TIME COMPLEXITY
Find      : O(log n)
Union     : O(log n)
Rollback  : O(1)
```

Connected (0,3): true

After Rollback

Connected (0,3): false

Time Complexity:

Find: $O(\log n)$

Union: $O(\log n)$

Rollback: $O(1)$

GITHUB LINK : <https://github.com/P-Shashe-Preetham/DSA-CASE-STUDIES.git>

Student Signature	Faculty Signature